



EXAMINATIONS COUNCIL OF ESWATINI
Eswatini General Certificate of Secondary Education

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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MATHEMATICS

6880/02

Paper 2 Structured Questions (Core)

October/November 2024

2 hours

Candidates answer on the Question Paper.

Additional Materials: Scientific calculator
Geometrical Instruments
Tracing paper (optional)

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name in the spaces provided.
Write in dark blue or black pen in the spaces provided on the Question Paper.
You may use a soft pencil for any diagrams or graphs.
Do **not** use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

All working should be clearly shown below that question.
The number of marks is given in brackets [] at the end of each question **or** part question.

Scientific calculators should be used.
If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures.
Give answers in degrees to one decimal place.
For π , use either your calculator value or 3.142.

The total of the marks for this paper is 90.

For Examiner's Use	
1	
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3	
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7	
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10	
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13	
Total	

This document consists of **16** printed pages.

- 1 (a) $\sqrt{2}$, 3, π , 8, 9, 18, 27, 36, 37

From the list of numbers, write down

- (i) all the prime numbers,

Answer (a)(i) [1]

- (ii) all the irrational numbers,

Answer (a)(ii) [1]

- (iii) all the cube numbers.

Answer (a)(iii) [1]

- (b) Arrange the following in order of size, starting with the **smallest**.

$\frac{3}{7}$, 42%, 0.426, $\sqrt{0.18}$, $\frac{5}{12}$

Answer (b),,,, [2]
smallest

- 2 Given that $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$,

$A = \{1, 2, 4, 7, 9, 10, 13, 15\}$ and $B = \{\text{Triangular numbers}\}$.

- (a) List all the elements of B .

Answer (a) [2]

- (b) Draw a Venn diagram to show the three sets.

[3]

- (c) Find $n(A \cap B')$.

Answer (c) [2]

- 3** The monthly income of a man is E15 714 before tax is deducted.

He pays E2750 in income tax.

- (a)** Calculate the percentage tax rate.

Answer (a) % [2]

- (b)** He then spends E2400 on groceries.

He spends E1600 on transport.

He spends E2000 on leisure.

He saves the remainder.

- (i)** Express the amount he spends on groceries as a fraction of his monthly income.

Give your answer in its simplest form.

Answer (b)(i) [2]

- (ii)** Calculate the amount he saves.

Answer (b)(ii) E [2]

- 4 A total of 40 students attended an agriculture lesson.

Each student was asked the number of fruit trees planted in the yard at their home.

The results of their responses are shown below.

3	7	2	3	1	4	2	6
4	3	2	4	7	2	2	4
5	2	3	2	4	3	2	4
4	2	7	2	4	2	4	1
7	6	4	3	2	5	2	3

- (a) Complete the frequency table below.

Number of Trees	Number of Students
1	2
2	13
3	
4	
5	
6	
7	

[2]

- (b) State the modal number of trees.

Answer (b) [1]

- (c) Find

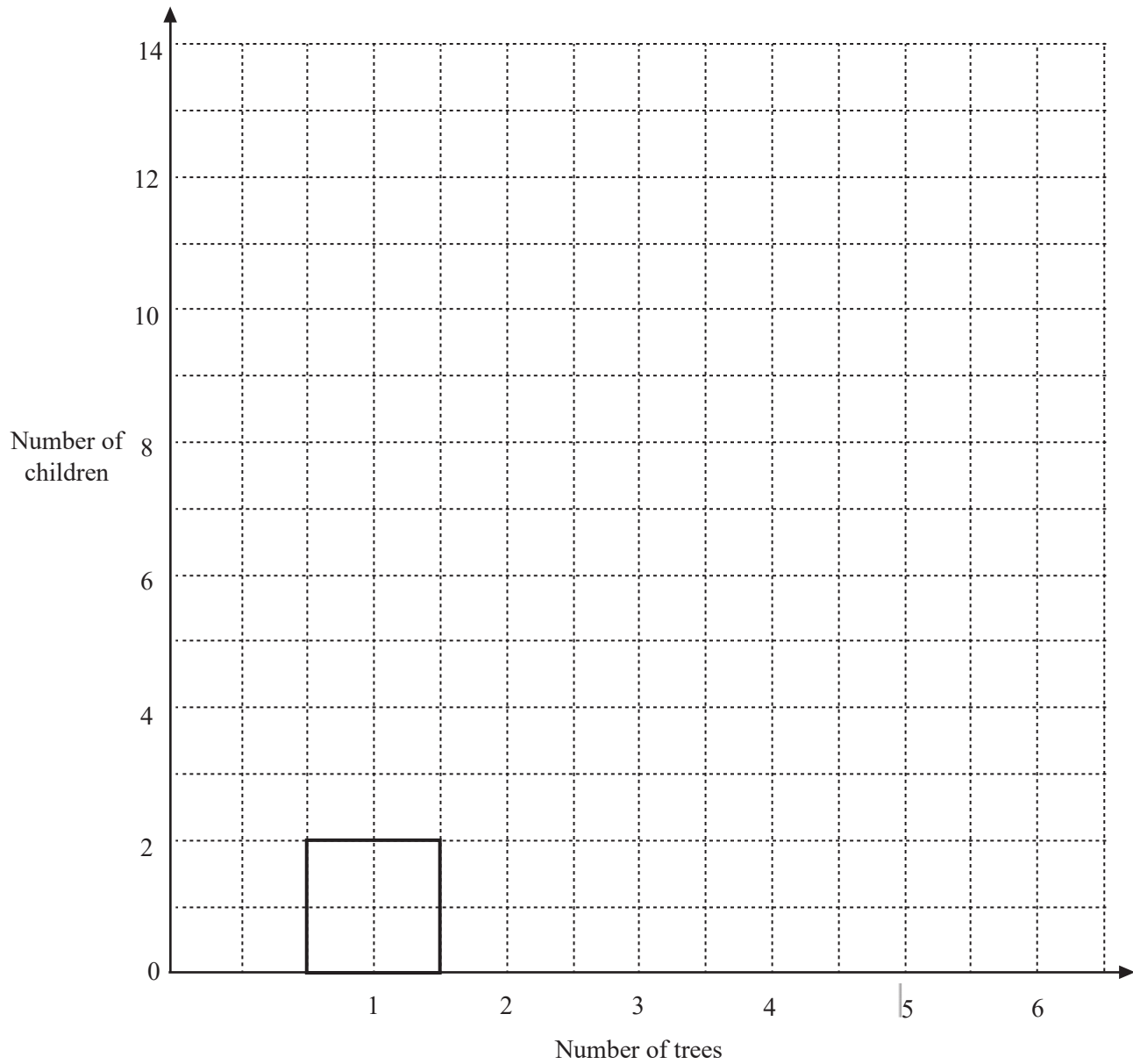
- (i) the range,

Answer (c)(i) [1]

- (ii) the median number of trees.

Answer (c)(ii) [2]

(d) Complete the bar chart to show the number of trees planted in the students' yard.

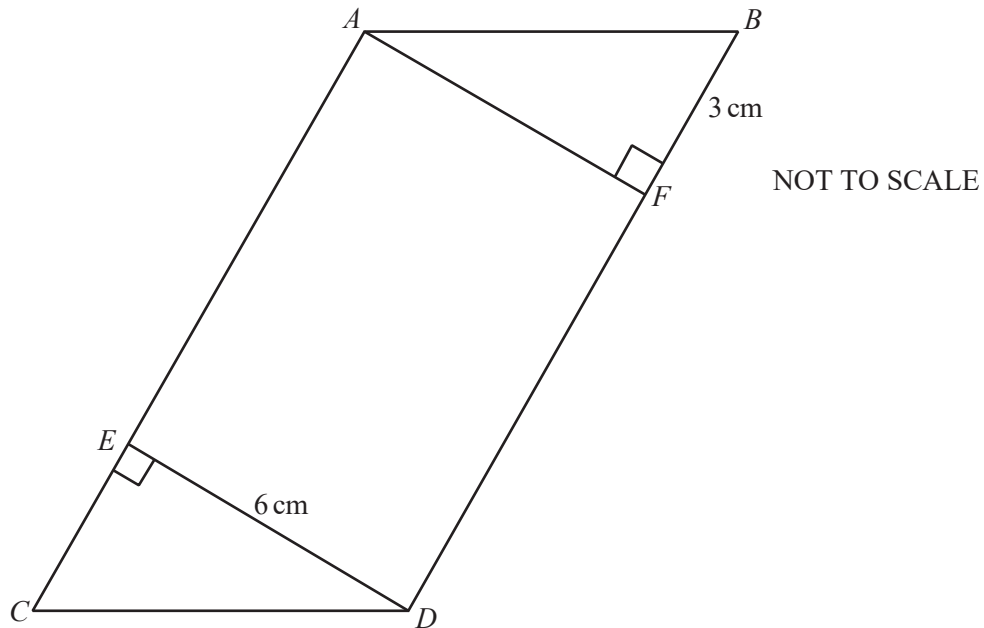


[2]

5 $ABDC$ is a parallelogram.

BAF and CDE are right angled triangles.

$BF = 3\text{ cm}$ and $DE = 6\text{ cm}$.



(a) Complete the sentence:

Triangle CDE is to triangle BAF . [1]

(b) Calculate the area of triangle CDE .

Answer (b) cm [2]

(c) Calculate angle ECD .

Answer (c) ° [2]

- 6 The table shows the times for Thembani Transport, travelling from Lomahasha to Manzini.

	Lomahasha		Simunye		Manzini	
Thembani Transport	Arrive	Depart	Arrive	Depart	Arrive	Depart
		0600 hrs	0625 hrs	0630 hrs	0730 hrs	

- (a) State the arrival time of Thembani Transport in Simunye.

Answer (a) [1]

- (b) Find the time taken by Thembani Transport to travel from Lomahasha to Manzini.

Answer (b) [1]

- (c) If the average speed of Thembani Transport from Lomahasha to Manzini is 60 km/hr, show that the distance between Lomahasha and Manzini is 90 km.

[2]

- (d) On the return journey the bus departed from Manzini at 1355 hours.

The average speed for the return journey was 75 km/h

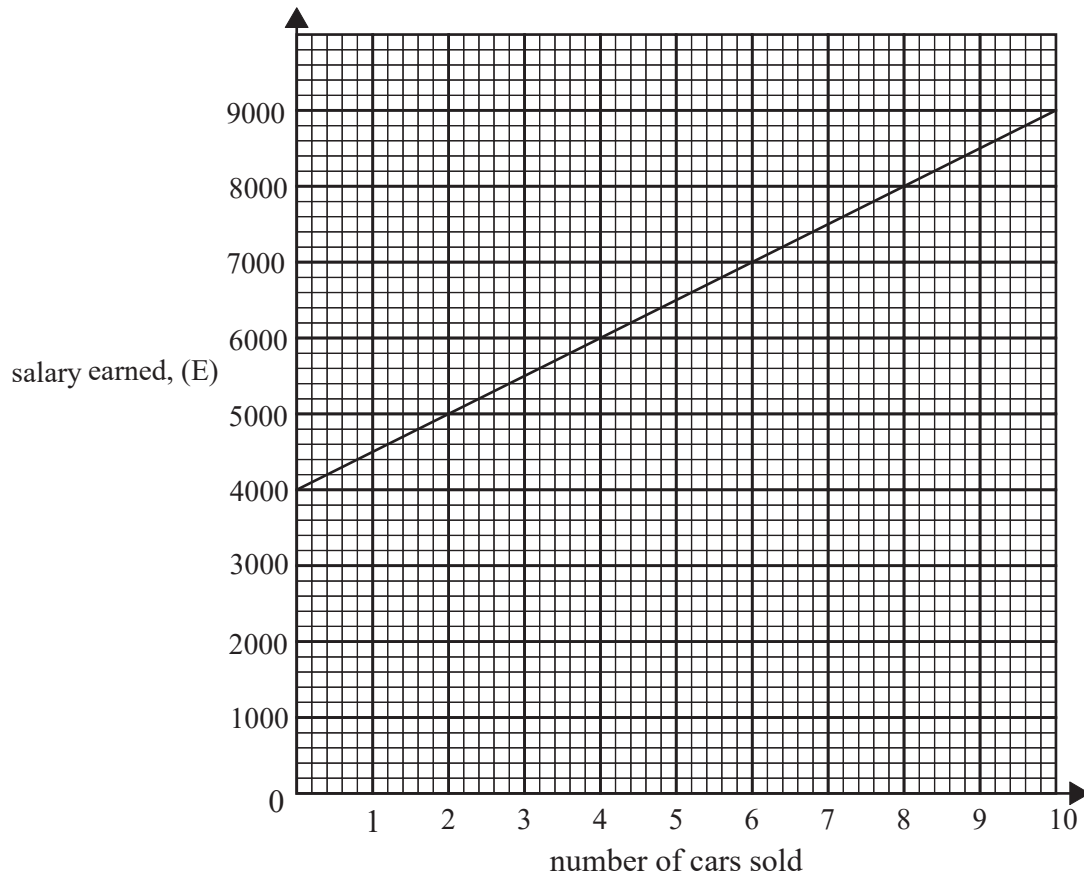
Find the time Thembani Transport arrived at Lomahasha.

Answer (d) [3]

- 7 The salary of a car salesman is made up of a fixed amount and a commission.

The commission is proportional to the number of cars he sells in a month.

The following is part of a graph that shows the salary the salesman can earn in a month.



- (a) Use the graph to find

- (i) the fixed amount of his salary per month,

Answer (a)(i) E [1]

- (ii) the salary earned when 4 cars are sold,

Answer (a)(ii) E [1]

- (iii) the gradient of the straight line.

Answer (a)(iii) [2]

- (b) What information is given by the gradient of the straight line in (a)(iii) above?

.....
 [1]

- (c) Find the salary earned if 24 cars are sold in a month.

Answer (c) E [2]

8 Solve.

(a) (i) $7 - x = 2(x - 1)$

Answer (a)(i) $x =$ [3]

(ii) $x^2 = 13x - 42$

Answer (a)(ii) $x =$ or $x =$ [4]

(b) Factorise $2pxy - 8x$.

Answer (b) [2]

- 9 Lomasontfo and Xolile were each given E5 as pocket money.

Both of them bought muffins, apples and sweets.

Each muffin cost E1.50, each apple cost 80 c and each sweet cost 20 c.

This information can be represented in matrix form as shown below.

$$(1.50 \quad 0.80 \quad 0.20)$$

- (a) State the order of the matrix shown above.

Answer (a) [1]

- (b) Lomasontfo bought 2 muffins, 1 apple and 1 sweet.

Xolile bought 1 muffin, 2 apples and 2 sweets.

This information can be represented in matrix form as shown below.

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 2 \end{pmatrix}$$

- (i) Work out

$$(1.50 \quad 0.80 \quad 0.20) \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 2 \end{pmatrix}$$

Answer (b)(i) [2]

- (ii) State the information represented by the matrix multiplication in (b)(i) above.

.....
..... [1]

- (iii) Find the amount of money Lomasontfo was left with.

Answer (b)(iii) E [1]

- 10** Athlete A and athlete B ran a relay in two days.

Each day the two athletes ran a combined distance of 50 km.

On Day 1 of the race, the athlete A ran a distance of x km.

- (a)** Write in terms of x the distance that was run by the athlete B on Day 1.

Answer (a) km [1]

- (b)** On Day 2 of the race the athlete A ran **half** the distance he ran on Day 1.

Write in terms of x the distance that was run by

- (i)** athlete A on Day 2 of the race,

Answer (b)(i) km [1]

- (ii)** athlete B on Day 2 of the race.

Answer (b)(ii) km [1]

- (c)** The distance run by athlete B on Day 2 of the race is three times the distance he ran on Day 1.

- (i)** Form an equation in terms of x

Answer (c)(i) [1]

- (ii)** Solve the equation formed in **(c)(i)** for x .

Answer (c)(ii) $x =$ [3]

- (iii)** Find the total distance covered by each athlete in the two days of the race.

Answer (c)(iii) athlete A km [1]

athlete B km [1]

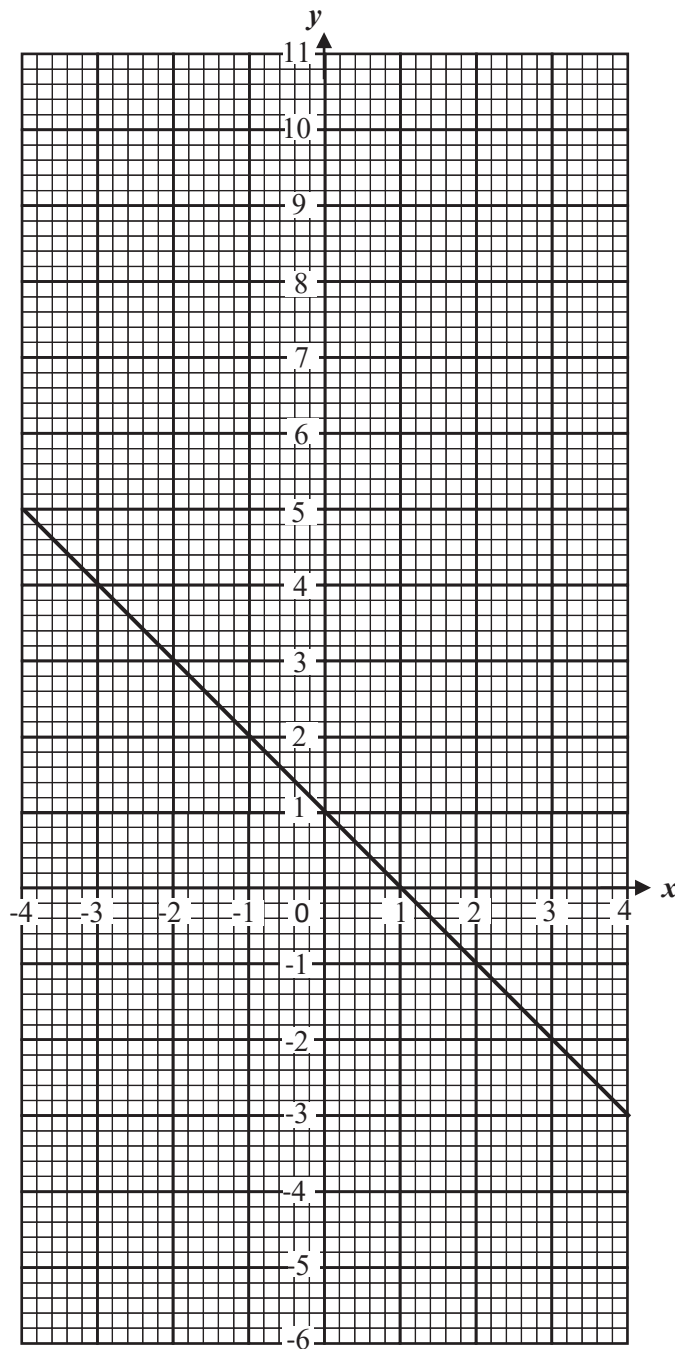
11 The grid shows a straight line with the equation $x + y = 1$

(a) The equation of a curve is $y = x^2 - 5$.

(i) Complete the table for the x and y values of the line $y = x^2 - 5$.

x	-4	-3	-2	-1	0	1	2	3	4
y	11	4		-4	-5	-4	-1		11

[2]



(ii) On the grid, draw the graph of $y = x^2 - 5$ for $-4 \leq x \leq 4$. [3]

(iii) Write down the coordinates of the points of intersection of the graphs of $x + y = 1$ and $y = x^2 - 5$.

Answer (a)(iii) (,) and (,) [2]

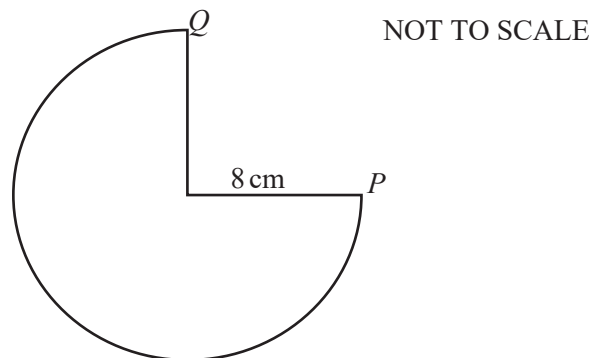
(b) Write the equation $4x + 3y = 6$ in the form $y = mx + c$.

Answer (b) [2]

- 12** A piece of paper is cut to form a net of a cone.

The net is a sector of a circle with radius 8 cm.

The sector angle is 270° .



- (a)** Calculate the area of the sector.

Answer (a) cm^2 [2]

- (b)** The paper is then folded to form the cone.

The length of the major arc PQ is 12π cm.

Calculate the radius of the cone.

Answer (b) cm [2]

- (c)** The height of the cone is $\sqrt{28}$ cm.

Calculate the volume of the cone.

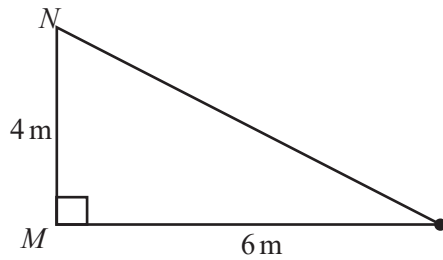
[Volume of a cone = $\frac{1}{3}\pi r^2 h$]

Answer (c) cm^3 [2]

- 13** A pole MN is held vertically on horizontal ground.

The pole is 4 m long.

A point P is 6 m away from the base of the pole.



NOT TO SCALE

- (a)** Calculate

- (i)** the angle of elevation of N from P ,

Answer (a)(i) [2]

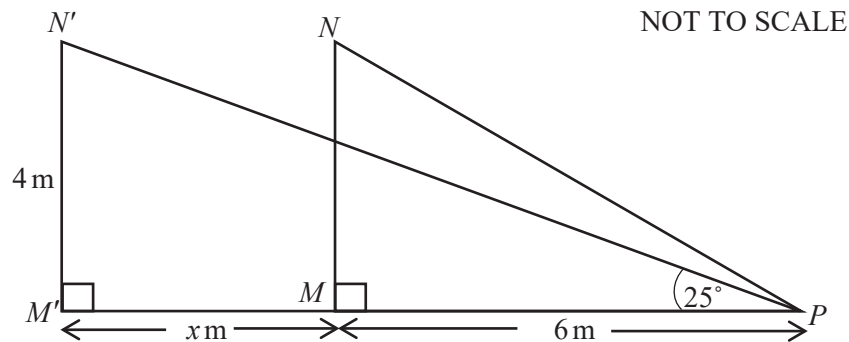
- (ii)** the distance PN .

Answer (a)(ii) m [2]

- (b) The pole is moved a distance x m from its original position.

The new position of N is N' and the new position of M is M' .

The angle of elevation of N' from P is 25° .



Find x .

Answer (b) $x = \dots\dots\dots$ [3]